

Welcome to the Other Side

How a social VR concert production broke the world's record of audiences across all media

Louis Cacciuttolo
VRrOOM

ACM Reference Format:

Louis Cacciuttolo. 2021. Welcome to the Other Side: How a social VR concert production broke the world's record of audiences across all media. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH '21 Immersive Pavilion)*, August 09–13, 2021, Virtual Event, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3450615.3464950>

1 ABSTRACT

“Virtual production has become indispensable for any choice of media distribution for physical events as well as for digital ones. This is the dawn of a brave new world for entertainment and arts. There is no coming back.”

Louis Cacciuttolo, CEO, VRrOOM



Last year, instead of putting on its traditional New Year's Eve celebration, the city of Paris commissioned Jean-Michel Jarre and the team of VRrOOM to create a virtual event to welcome in 2021, with viewers able to watch from the safety of their home. According to Sony Music International, the event has now garnered more than 75 million views across social media platforms including Facebook, VRC, Weibo and TikTok, as well as various TV channels and radio stations. “When it comes to virtual shows, it's not the amazing effects or impressive CGI that keeps an audience's attention – it has to be engaging, otherwise you'll switch off after 15 minutes. There is a reason why people stay up for hours playing a rather rudimentary 8-bit video game. It's because we were participating in something,” says Jvan Morandi, the concert's scenographer.

The concert lasted 55 minutes and took viewers into a virtual experience with a digital replica of Notre Dame Cathedral. The performance could be viewed via a VR headset in which gig-goers

were able to move around the virtual Notre Dame and interact with the surroundings. Alternatively, viewers could tune into a live cut of the virtual show, which spliced real-time footage of the artist in his studio with an animated JMJ avatar performing within the 3D model of the Cathedral and Notre Dame.

Before working out how to create a show within the gaming engine, Jvan Morandi, Jean-Michel Jarre long-time's scenographer, and the VRrOOM team went through a number of conceptual ideas, all of which were designed on Capture. Morandi joined forces with the Producer, Louis Cacciuttolo and 3D Animation Artist, Vincent Masson of VMD. 3D Modelling and Unity Game Engine specialists, Lapo Germasi and Victor Pukhov from Manifattura Italiana Design, alongside Unity Scripting and Lead Developer Antony Vitillo were key to the creation of the show.

“The team from VRrOOM have a huge amount of experience when it comes to VR and were incredibly helpful converting the ideas developed in Capture and bringing them to the Unity Engine. Despite working in this new format, the team was able to use some typical tools of the trade in the creation of the show – namely a ChamSys MQ80. All the cue lists came from ChamSys and were translated into a set of Unity animation triggers on a timeline. ‘Translated’ means that Victor Pukhov used the visualized lighting cues to create ‘shader animations’ that then got triggered by Unity custom scripts by Antony Vitillo.

The basic principles of stage and show design apply to creating a show in the virtual realm along with some notable differences – specifically when treating lighting effects. In a gaming engine, stage light beams are not real light but volumetric shaders, 3D objects with a texture. This means to create any big lighting looks, Pukhov from Manifattura Italiana Design had to mimic any moves which Morandi had programmed with the ChamSys desk. This whole project has all hinged on finding ways of clear communication, which is not easy because of often different workflow styles between show business and game development.

360° VR Camera Director Georgy Molodtsov, VR Camera Director and Editor Maud Clavier and Live Broadcast Director David Montagne filmed the whole social VR concert in 2D with a system developed by VRrOOM of virtual cameras working inside VRChat.

A number of creatives were dotted around the UK and Italy, while Georgy Molodtsov operated from Moscow along with Maud in Paris. Each director oversaw up to eight remote VR cameras and drones within the virtual world.

Responsible for inserting Jarre into the virtual world was TV Studio Gabriel, Paris, where the artist performance was translated live with motion capture technology and then mirrored by an in performance avatar that had been created by Freddy Kone and Mo Marouene of SoWhen. Morandi believes that if this type of show was to exist alongside, say, a tour, then the speed at which a VR show would be created would have to match up with that of a live tour.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '21 Immersive Pavilion, August 09–13, 2021, Virtual Event, USA

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8368-4/21/08.

<https://doi.org/10.1145/3450615.3464950>



“One of the interesting differences we have in live touring compared to those working within the gaming industry is timescales.

A regular show might take between three to six months to design, whereas video games by their very nature can take up to seven years to design.

Most companies behind the major gaming engines do not know the world of live events, says Jvan Morandi. Right now, we are seeing a mirroring of what happened with the record industry decades ago. You look now and the major streaming companies – Spotify and Apple – have not come from the music industry.”

ACKNOWLEDGMENTS

With the support of Ville de Paris, Paris Mayor Anne Hidalgo, le Diocèse, Mgr. Aupetit, Mgr. Chauvet, and UNESCO.